

Esercizio 1. Calcolare la risposta nell'uscita del sistema

$$\begin{aligned}x(t+1) &= Ax(t) + Bu(t) \\ y(t) &= Cx(t) + Du(t)\end{aligned}$$

con

$$\begin{aligned}A &= \begin{pmatrix} 2 & 0 & 1 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{pmatrix} \\ B &= \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix} \\ C &= (1 \quad -1 \quad 0) \\ D &= 3 \\ u(t) &= \sin(2t) \delta_{-1}(t) \\ x(0) &= \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}\end{aligned}$$

Utilizzo la trasformata Z

$$\begin{aligned}u(z) &= \mathcal{Z}[u(t)] \\ &= \frac{z \sin(2)}{z^2 - 2 \cos(2)z + 1}\end{aligned}$$

$$\begin{aligned}(zI - A)^{-1} &= \left(\begin{array}{c|cc} z-2 & 0 & -1 \\ 0 & z-2 & 0 \\ 0 & 0 & z-2 \end{array} \right)^{-1} \\ &= \frac{1}{(z-2)^3} \begin{pmatrix} (z-2)^2 & 0 & (z-2) \\ 0 & (z-2)^2 & 0 \\ 0 & 0 & (z-2)^2 \end{pmatrix} \\ &= \begin{pmatrix} \frac{1}{z-2} & 0 & \frac{1}{(z-2)^2} \\ 0 & \frac{1}{z-2} & 0 \\ 0 & 0 & \frac{1}{z-2} \end{pmatrix} \\ &= \left(\begin{array}{c|cc} \frac{1}{z-2} & 0 & \frac{1}{(z-2)^2} \\ 0 & \frac{1}{z-2} & 0 \\ 0 & 0 & \frac{1}{z-2} \end{array} \right)\end{aligned}$$

$$\begin{aligned}\left(\begin{array}{c|c} A & B \\ \hline 0 & C \end{array} \right)^{-1} &= \left(\begin{array}{c|c} A^{-1} & X \\ \hline 0 & C^{-1} \end{array} \right) \\ \left(\begin{array}{c|c} A^{-1} & X \\ \hline 0 & C^{-1} \end{array} \right) \cdot \left(\begin{array}{c|c} A & B \\ \hline 0 & C \end{array} \right) &= \left(\begin{array}{c|c} I & 0 \\ \hline 0 & I \end{array} \right) \\ A^{-1}B + XC &= 0 \\ X &= -A^{-1}BC^{-1}\end{aligned}$$

$$\begin{aligned}
&= -\frac{1}{z-2} \begin{pmatrix} 0 & -1 \end{pmatrix} \begin{pmatrix} \frac{1}{z-2} & 0 \\ 0 & \frac{1}{z-2} \end{pmatrix} \\
&= \begin{pmatrix} 0 & \frac{1}{(z-2)^2} \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
\left(\begin{array}{c|c} A & 0 \\ \hline B & C \end{array} \right)^{-1} &= \left(\begin{array}{c|c} A^{-1} & 0 \\ \hline X & C^{-1} \end{array} \right) \\
\left(\begin{array}{c|c} A^{-1} & 0 \\ \hline X & C^{-1} \end{array} \right) \left(\begin{array}{c|c} A & 0 \\ \hline B & C \end{array} \right) &= \left(\begin{array}{c|c} I & 0 \\ \hline 0 & I \end{array} \right) \\
XA + C^{-1}B &= 0 \\
X &= -C^{-1}BA^{-1}
\end{aligned}$$

$$\begin{aligned}
H(z) &= zC(zI - A)^{-1} \\
&= z \begin{pmatrix} 1 & -1 & 0 \end{pmatrix} \cdot \begin{pmatrix} \frac{1}{z-2} & 0 & \frac{1}{(z-2)^2} \\ 0 & \frac{1}{z-2} & 0 \\ 0 & 0 & \frac{1}{z-2} \end{pmatrix} \\
&= \begin{pmatrix} \frac{z}{z-2} & -\left(\frac{z}{z-2}\right) & \frac{z}{(z-2)^2} \end{pmatrix} \\
y_l(z) &= H(z)x_0 \\
&= \begin{pmatrix} \frac{z}{z-2} & -\left(\frac{z}{z-2}\right) & \frac{z}{(z-2)^2} \end{pmatrix} \cdot \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \\
&= \frac{z}{(z-2)^2} \\
y_l(t) &= \binom{t}{1} 2^{t-1}
\end{aligned}$$

$$\begin{aligned}
W(z) &= C(zI - A)^{-1}B + D \\
&= \begin{pmatrix} 1 & -1 & 0 \end{pmatrix} \cdot \begin{pmatrix} \frac{1}{z-2} & 0 & \frac{1}{(z-2)^2} \\ 0 & \frac{1}{z-2} & 0 \\ 0 & 0 & \frac{1}{z-2} \end{pmatrix} \cdot \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix} + 3 \\
&= -\left(\frac{1}{z-2}\right) + \frac{1}{(z-2)^2} + 3 \\
y_f(z) &= W(z)u(z) \\
&= \left(-\left(\frac{1}{z-2}\right) + \frac{1}{(z-2)^2} + 3\right) \frac{z \sin(2)}{z^2 - 2 \cos(2)z + 1}
\end{aligned}$$

Consideriamo i singoli termini

$$\begin{aligned}
z \left(-\left(\frac{1}{z-2}\right) \frac{\sin(2)}{z^2 - 2 \cos(2)z + 1} \right) &= z \left(\frac{A}{z-2} + B \frac{\sin(2)}{z^2 - 2 \cos(2)z + 1} + C \frac{z - \cos(2)}{z^2 - 2 \cos(2)z + 1} \right) \\
&= z \left(\frac{\bar{A}}{z-2} + \frac{\bar{B}z + \bar{C}}{z^2 - 2 \cos(2)z + 1} \right)
\end{aligned}$$

$$\begin{aligned}
A &= \text{subst}\left(z=2, -\left(\frac{1}{1}\right) \frac{\sin(2)}{z^2 - 2\cos(2)z + 1}\right) \\
&= -\left(\frac{\sin(2)}{5 - 4\cos(2)}\right) \\
0 &= A + C \\
C &= -A \\
&= \left(\frac{\sin(2)}{5 - 4\cos(2)}\right) \\
\frac{\sin(2)}{2} &= -(\cos(2))C + \sin(2)B - \frac{A}{2} \\
\sin(2)B &= \frac{\sin(2)}{2} + (\cos(2))C + \frac{A}{2} \\
&= \frac{\sin(2)}{2} + (\cos(2))\left(\frac{\sin(2)}{5 - 4\cos(2)}\right) - \frac{1}{2}\left(\frac{\sin(2)}{5 - 4\cos(2)}\right) \\
&= \frac{(\cos(2) - 2)\sin(2)}{4\cos(2) - 5} \\
y_{f,1}(t) &= z\left(\frac{A}{z-2} + B\frac{\sin(2)}{z^2 - 2\cos(2)z + 1} + C\frac{z - \cos(2)}{z^2 - 2\cos(2)z + 1}\right) \\
&= A2^t + B\sin(2t) + C\cos(2t) \\
z\left(\frac{1}{(z-2)^2} \frac{\sin(2)}{z^2 - 2\cos(2)z + 1}\right) &= z\left(\frac{A}{z-2} + \frac{B}{(z-2)^2} + C\frac{\sin(2)}{z^2 - 2\cos(2)z + 1} + \right. \\
&\quad \left. D\frac{z - \cos(2)}{z^2 - 2\cos(2)z + 1}\right)
\end{aligned}$$

Esercizio 2. Calcolare una realizzazione della seguente funzione di trasferimento

$$W(s) = \frac{s^2 + 2s + 3}{s^2 + s + 1}$$

$$\begin{aligned}
D &= \lim_{s \rightarrow \infty} W(s) \\
&= 1 \\
V(s) &= W(s) - D \\
&= \frac{s^2 + 2s + 3}{s^2 + s + 1} - 1 \\
&= \frac{s + 2}{s^2 + s + 1}
\end{aligned}$$

$$A_c = \begin{pmatrix} 0 & 1 \\ -1 & -1 \end{pmatrix}$$

$$B_c = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

$$C_c = (2 \ 1)$$

$$D_c = 1$$

$$A_o = \begin{pmatrix} 0 & -1 \\ 1 & -1 \end{pmatrix}$$

$$B_o = \begin{pmatrix} 2 \\ 1 \end{pmatrix}$$

$$\begin{aligned}C_o &= \begin{pmatrix} 0 & 1 \end{pmatrix} \\D_o &= 1\end{aligned}$$

$$\begin{aligned}W(s) &= \begin{pmatrix} 2 & 1 \end{pmatrix} \cdot \begin{pmatrix} \frac{s+1}{s(s+1)+1} & \frac{1}{s(s+1)+1} \\ -\left(\frac{1}{s(s+1)+1}\right) & \frac{s}{s(s+1)+1} \end{pmatrix} \cdot \begin{pmatrix} 0 \\ 1 \end{pmatrix} + 1 \\&= \frac{s+2}{s^2+s+1} + 1 \\&= \frac{s^2+2s+3}{s^2+s+1}\end{aligned}$$

Esercizio 3. Si valuti osservabilità ed eccitabilità modale del sistema con le seguenti matrici

$$\begin{aligned}A &= \begin{pmatrix} 1 & 2 & 0 \\ 0 & -1 & 1 \\ 0 & 0 & -2 \end{pmatrix} \\B &= \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix} \\C &= \begin{pmatrix} 1 & 0 & 2 \end{pmatrix} \\D &= 1\end{aligned}$$

Opero un cambio di base per portare A in forma diagonale

$$\begin{aligned}p(s) &= \det(sI - A) \\&= \det \begin{pmatrix} s-1 & -2 & 0 \\ 0 & s+1 & -1 \\ 0 & 0 & s+2 \end{pmatrix} \\&= (s-1)(s+1)(s+2) \\ \sigma(A) &= \{1, -1, -2\}\end{aligned}$$

$$\begin{aligned}\lambda_1 &= 1 \\ \text{Ker}(\lambda_1 I - A) &= \text{Ker} \begin{pmatrix} 0 & -2 & 0 \\ 0 & 2 & -1 \\ 0 & 0 & 3 \end{pmatrix} \\&= \text{span} \left(\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} \right)\end{aligned}$$

$$\begin{aligned}\lambda_2 &= -1 \\ \text{Ker}(\lambda_2 I - A) &= \begin{pmatrix} -2 & -2 & 0 \\ 0 & 0 & -1 \\ 0 & 0 & 1 \end{pmatrix} \\&= \text{span} \left(\begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix} \right)\end{aligned}$$

$$\begin{aligned}\lambda_3 &= -2 \\ \text{Ker}(\lambda_3 I - A) &= \begin{pmatrix} -3 & -2 & 0 \\ 0 & -1 & -1 \\ 0 & 0 & 0 \end{pmatrix} \\ &= \text{span}\left(\begin{pmatrix} -\frac{2}{3} \\ 1 \\ -1 \end{pmatrix}\right)\end{aligned}$$

$$T_1^{-1} = \begin{pmatrix} -\frac{2}{3} & 1 & 1 \\ 1 & 0 & -1 \\ -1 & 0 & 0 \end{pmatrix} \text{ non ha forma triangolare (scelta sbagliata)}$$

$$T_1 = \begin{pmatrix} 0 & 0 & -1 \\ 1 & 1 & \frac{1}{3} \\ 0 & -1 & -1 \end{pmatrix}$$

$$\begin{aligned}T_1 A T_1^{-1} &= \begin{pmatrix} 0 & 0 & -1 \\ 1 & 1 & \frac{1}{3} \\ 0 & -1 & -1 \end{pmatrix} \cdot \begin{pmatrix} 1 & 2 & 0 \\ 0 & -1 & 1 \\ 0 & 0 & -2 \end{pmatrix} \cdot \begin{pmatrix} -\frac{2}{3} & 1 & 1 \\ 1 & 0 & -1 \\ -1 & 0 & 0 \end{pmatrix} \\ &= \begin{pmatrix} -2 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{pmatrix}\end{aligned}$$

$$T^{-1} = \left(\begin{array}{c|cc} 1 & 1 & -\frac{2}{3} \\ \hline 0 & -1 & 1 \\ 0 & 0 & -1 \end{array} \right) \text{ ha forma triangolare! (buono)}$$

$$T = \left(\begin{array}{c|cc} 1 & 1 & \frac{1}{3} \\ \hline 0 & -1 & -1 \\ 0 & 0 & -1 \end{array} \right)$$

$$\begin{aligned}T T^{-1} &= \left(\begin{array}{c|cc} 1 & 1 & \frac{1}{3} \\ \hline 0 & -1 & -1 \\ 0 & 0 & -1 \end{array} \right) \cdot \left(\begin{array}{c|cc} 1 & 1 & -\frac{2}{3} \\ \hline 0 & -1 & 1 \\ 0 & 0 & -1 \end{array} \right) \\ &= \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}\end{aligned}$$

$$\begin{aligned}X &= -1 \left(\begin{array}{c|c} 1 & -\frac{2}{3} \\ \hline 1 & -\frac{2}{3} \end{array} \right) \cdot \left(\begin{array}{c|c} -1 & -1 \\ \hline 0 & -1 \end{array} \right) \\ &= \begin{pmatrix} 1 & \frac{1}{3} \end{pmatrix}\end{aligned}$$

$$z = Tx$$

$$\bar{A} = T A T^{-1}$$

$$= \left(\begin{array}{c|cc} 1 & 1 & \frac{1}{3} \\ \hline 0 & -1 & -1 \\ 0 & 0 & -1 \end{array} \right) \cdot \begin{pmatrix} 1 & 2 & 0 \\ 0 & -1 & 1 \\ 0 & 0 & -2 \end{pmatrix} \cdot \left(\begin{array}{c|cc} 1 & 1 & -\frac{2}{3} \\ \hline 0 & -1 & 1 \\ 0 & 0 & -1 \end{array} \right)$$

$$\begin{aligned}
&= \begin{pmatrix} \textcolor{red}{1} & 0 & 0 \\ 0 & \textcolor{blue}{-1} & 0 \\ 0 & 0 & \textcolor{green}{-2} \end{pmatrix} \\
\bar{B} &= TB \\
&= \left(\begin{array}{c|cc} 1 & 1 & \frac{1}{3} \\ 0 & -1 & -1 \\ 0 & 0 & -1 \end{array} \right) \cdot \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix} \\
&= \begin{pmatrix} \textcolor{red}{\frac{4}{3}} \\ \textcolor{blue}{-2} \\ \textcolor{green}{-1} \end{pmatrix} \quad \text{modi } \textcolor{red}{e}^t, \textcolor{blue}{e}^{-t}, \textcolor{green}{e}^{-2t} \text{ sono eccitabili dall'ingresso} \\
\bar{C} &= CT^{-1} \\
&= (1 \ 0 \ 2) \cdot \left(\begin{array}{c|cc} 1 & 1 & -\frac{2}{3} \\ 0 & -1 & 1 \\ 0 & 0 & -1 \end{array} \right) \\
&= \left(\textcolor{red}{1} \ \textcolor{blue}{1} \ -\left(\frac{8}{3}\right) \right) \quad \text{modi } \textcolor{red}{e}^t, \textcolor{blue}{e}^{-t}, \textcolor{green}{e}^{-2t} \text{ sono osservabili dall'uscita}
\end{aligned}$$

Esercizio 4. Si calcoli una realizzazione della seguente funzione di trasferimento

$$\begin{aligned}
W(z) &= \frac{z^2+1}{z^2-1} \\
D &= \lim_{z \rightarrow \infty} W(z) \\
&= 1 \\
V(z) &= W(z) - D \\
&= \frac{z^2+1}{z^2-1} - 1 \\
&= \frac{2}{z^2-1} \\
&= \text{partfrac}\left(\frac{2}{z^2-1}, z\right) \\
&= \frac{\textcolor{red}{1}}{\textcolor{red}{z}-\textcolor{red}{1}} + \frac{\textcolor{blue}{-1}}{\textcolor{blue}{z}+\textcolor{blue}{1}} \\
A_c &= \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \\
B_c &= \begin{pmatrix} 0 \\ 1 \end{pmatrix} \\
C_c &= (2 \ 0) \\
D_c &= 1 \\
A_o &= \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \\
B_o &= \begin{pmatrix} 2 \\ 0 \end{pmatrix} \\
C_o &= (0 \ 1) \\
D_o &= 1
\end{aligned}$$

$$A_f = \begin{pmatrix} \textcolor{red}{1} & 0 \\ 0 & \textcolor{blue}{-1} \end{pmatrix}$$

$$B_f = \begin{pmatrix} \textcolor{red}{1} \\ \textcolor{blue}{-1} \end{pmatrix}$$

$$C_f = (\textcolor{red}{1} \textcolor{blue}{1})$$

$$D_f = 1$$